

Effect of curing temperature on the development of hard structure of metakaolin-based geopolymer

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ABSTRACT

The properties of metakaolin-based geopolymer are directly impacted not only by the specific surface and composition of initial metakaolin and the type, composition and relative amount of alkali activator used but they also depend on the conditions during the initial period of geopolymerization reaction. This study aimed to analyze the effect of curing temperature (10, 20, 40, 60 and 80 °C) and time on the compressive and flexural strengths, pore distribution and microstructure of alkali activated metakaolin material. The results have shown that the treatment of fresh mixture at elevated temperatures accelerates the strengths development but the 28 days mechanical properties are deteriorated in comparison with results obtained for mixtures that were treated at an ambient or slightly decreased temperature. The influence of curing temperature on microstructure of geopolymer matrix was verified in terms of pore distribution studied by means of mercury intrusion porosimetry. The study revealed a tendency to increase pore size and cumulative pore volume with rising temperature, which is reflected in mechanical properties. It has been also shown the possibility of monitoring the geopolymerization reaction by means of Infrared Spectroscopy.

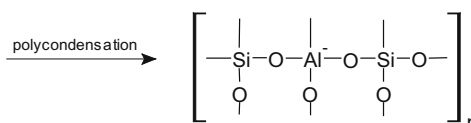
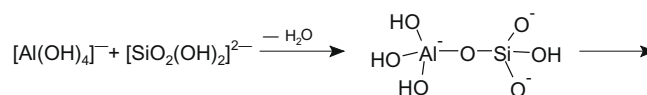
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1. Introduction

Since the late seventies of the 20th century geopolymers have been considered for replacing traditional structural materials by reason of their excellent properties and high performance [1,2]. During the last decade, increased research efforts have been directed to this area due to the wide range of potential applications of these materials besides civil engineering in many other branches of industry. Geopolymers belong to the group of strong and durable cementitious materials that harden at temperatures below 100 °C [3–5]. It is three-dimensional CaO-free aluminosilicate binder, which is usually prepared by alkali activation of metakaolin or other inorganic material having pozzolanic properties such as fly ash and some of the aluminosilicate-based natural minerals [6–8]. The reason for using mainly metakaolin to produce geopolymers could be the fact that it is common industrial mineral which can be obtained in a large quantity with homogeneous properties. Metakaolin is also environmentally friendly compared to Portland cement: its production requires much lower calcining temperature and emits 80–90% less CO₂ than Portland cement [9].

Alkali activation of metakaolin can be performed by solution of alkali hydroxide [10] or by alkali salt that gives after hydrolysis strongly basic solution, e.g. alkali silicate [11]. The process com-

prises dissolution of primary aluminosilicate framework from metakaolin followed by condensation of free silicate and aluminate species to form three-dimensional structure. This structure consists of cross-linked SiO₄ and AlO₄[−] tetrahedra where the negative charge on Al³⁺ in IV-fold coordination is balanced with positive charge of alkali ion (Na⁺, K⁺) [9]. The geopolymerization reaction can be expressed according to the following scheme:



This structure is very closed to the structure of zeolites but without regular ordering to longer distance – it has amorphous character. Long range ordering of tetrahedra is affected by temperature at which the polycondensation reaction occurs. During long term hydrothermal conditions at temperatures above 85 °C the

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formation of crystalline structure is preferred [3,12,13]. The reaction between metakaolin and alkaline solution has been studied by several analytical methods, among others by calorimetric measurements [14,15], thermal analysis (DTA, TGA and DSC) [16–18], FTIR and NMR [19–22] and X-ray diffraction analysis [13,14,23]. It has been proved that metakaolin-based geopolymer is highly amorphous material and the only crystalline phases that can be found in XRD patterns were assigned to quartz traces that were already present in metakaolin precursor.

The mechanical properties and microstructure of geopolymer strongly depends on the initial Si/Al ratio. Better strength proper-

ties have been reported for mixtures with $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratios in the range of 3.0–3.8 with an $\text{Na}_2\text{O}/\text{Al}_2\text{O}_3$ ratio of about one that can be easily achieved when sodium silicate with appropriate $\text{SiO}_2/\text{Na}_2\text{O}$ ratio is used. Changes in the $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio beyond this range usually result in low strength systems [24]. However, several results that have been published are not easy to compare because of different conditions, such as temperature, curing time, relative humidity, at which geopolymerization was carried out. Unfortunately, to date little is known about the influence of curing temperature on the mechanical properties and microstructure of geopolymer binder, which is supposed to be one of the important factors affecting the rate of formation and quality of the hard structure. Such information would be worth for instance in production of prefabricated elements made of composite that would use geopolymer as a binder.

This paper reports on the mechanical properties of metakaolin-based geopolymer mortars that were synthesized in different curing conditions concerning temperatures from 10 to 80 °C and different time of curing. The results of compressive and flexural strengths were explained from the viewpoint of microstructure changes that were determined by means of mercury intrusion porosimetry and FTIR spectroscopy during hardening process. The composition of the investigated geopolymer had been chosen on the basis of previous experience with several types of geopolymer materials prepared at an ambient temperature as the mixture having the best mechanical properties after 28 days.

Table 1

Chemical composition of starting materials.

Component (%)	Metakaolin	Alkaline silicate solution
Al_2O_3	40.94	–
SiO_2	55.01	24.9
K_2O	0.60	–
Na_2O	0.09	18.5
CaO	0.14	–
MgO	0.34	–
Fe_2O_3	0.55	–
TiO_2	0.55	–
LOI	1.54	–
H_2O	–	56.6

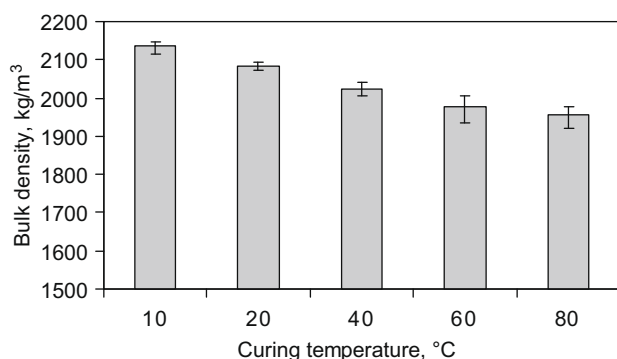
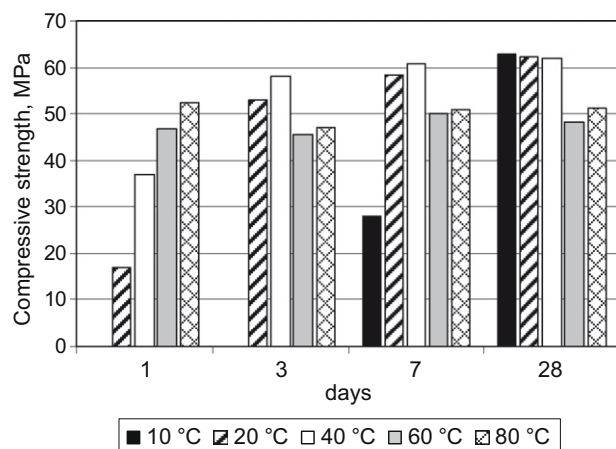


Fig. 1. Bulk density of geopolymer material prepared at different curing temperatures (10, 20, 40, 60 and 80 °C) at the age of 28 days. Heated mixtures were treated at elevated temperatures for initial 4 h. Error bars indicate the minimum and maximum value in the measurement series.



2. Experimental methods

2.1. Materials

Metakaolin was purchased from ČLUZ (CZ) under the brand name of Mefisto K05 that was produced by calcination of kaolin at 750 °C in rotary kiln. The molar composition of metakaolin determined by X-ray fluorescence is presented in Table 1. The Brunauer–Emmett–Teller (BET) surface area [25] of metakaolin determined by nitrogen absorption is 13.1 m²/kg, and the mean particle size (d_{50}) is 4.82 μm.

Alkaline silicate solution with silicate modulus ($\text{SiO}_2/\text{Na}_2\text{O}$) 1.39 was prepared by dissolving of solid sodium hydroxide (Lachema, 98.0%) in commercial sodium water glass (Zakłady chemiczne Rudniki, S.A., $\text{SiO}_2/\text{Na}_2\text{O}$ = 3.26 and $\text{H}_2\text{O}/\text{Na}_2\text{O}$ = 10.40) until clear. Standard quartz sand was added as an aggregate for the preparation of geopolymer mortars that were used for testing of mechanical properties. The maximum grain size of quartz was 2.5 mm.

2.2. Sample preparation

Geopolymer samples were prepared by mechanically mixing metakaolin and activator solution in planetary mixer for 5 min. Then quartz sand was mixed into this geopolymer paste with some additional water in order to obtain better workability of the mortar. The geopolymer mortar composed of 450 g metakaolin, 372 g

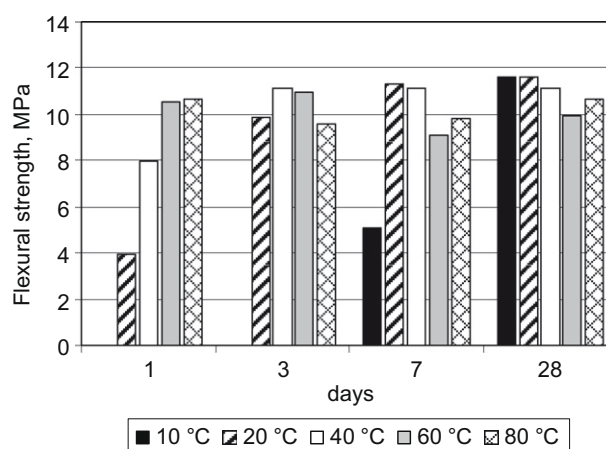


Fig. 2. Development of compressive and flexural strength of geopolymer material cured at 10, 20, 40, 60 and 80 °C over time (1, 3, 7, and 28 days). Curing at 40, 60 and 80 °C was carried out for initial 4 h.

alkaline silicate solution, 1350 g quartz sand, and 100 ml water. The slurry was cast into prismatic moulds with dimensions $40 \times 40 \times 160$ mm [26], vibrated for 2 min to remove entrained air and sealed. Prepared mixtures were cured at different curing regimes. In the first experiment, the specimens were cured at temperatures from 10 to 80 °C. One set of specimens was stored at an ambient temperature (20 °C) and was considered a reference material, one was stored in the fridge (10 °C) for the whole period before testing, and other specimens were cured at temperature 40, 60 or 80 °C in an electrical oven immediately after casting. After 4 h they were removed from oven and stored at an ambient temperature (20 °C) and relative humidity $45 \pm 5\%$ until tested. The second experiment was focused on the effect of different curing times at elevated temperatures. The specimens were treated for 1, 2, 3 and 4 h at temperatures 40, 60 or 80 °C. After the initial curing they were stored at an ambient temperature until tested. For FTIR and mercury intrusion porosimetry (MIP) analysis pure geopolymer paste without aggregate was prepared and transferred into small moulds ($20 \times 20 \times 100$ mm). Curing of these specimens followed the same procedures that were applied to larger specimens of geopolymer mortars. The reason for use of pure paste in FTIR analysis was the elimination of bands that belong to vibration modes of aggregate, which coincide with bands of geopolymer binder. In the case of porosimetry measurement, analysis of pure geo-

polymer paste eliminates the heterogeneity of mortar and influence of aggregate, it also gives better idea of the effect of temperature on pore structure of binder and provides better reproducibility and accuracy of the measurement. To assure the accuracy of the measurement, preparation of the samples followed always the same procedure. Specimens of geopolymer pastes were dried under vacuum at an ambient temperature in order to avoid further acceleration of geopolymerisation reaction that would probably occur in the case of heat drying in an oven. The samples of size $10 \times 10 \times 5$ mm that were used for direct measurements were cut with diamond micro saw Micron 110 (MTH Hrazdil Ltd.) having the constant load of 10 N and disc speed 2300 rpm.

2.3. Characterization

Bulk density was measured in accordance with EN 12390-7 at 28 days of curing. Mechanical properties were conducted on specimens made of geopolymer mortars with quartz sand at the age of 1, 3, 7 and 28 days. Flexural strengths were determined using standard three-point-bending test and compressive strengths were measured on far edge of both residual pieces obtained from flexural test according

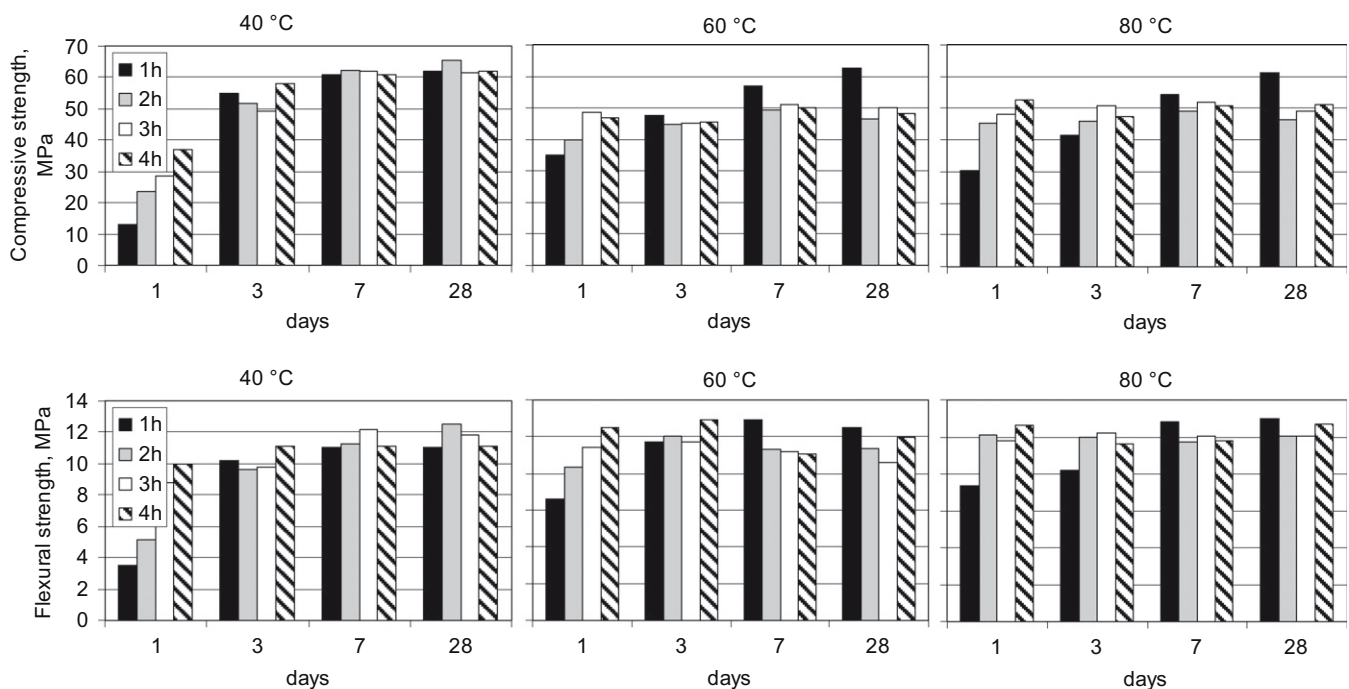


Fig. 3. Influence of curing time on compressive and flexural strengths of geopolymer cured at elevated temperatures (40, 60 and 80 °C).

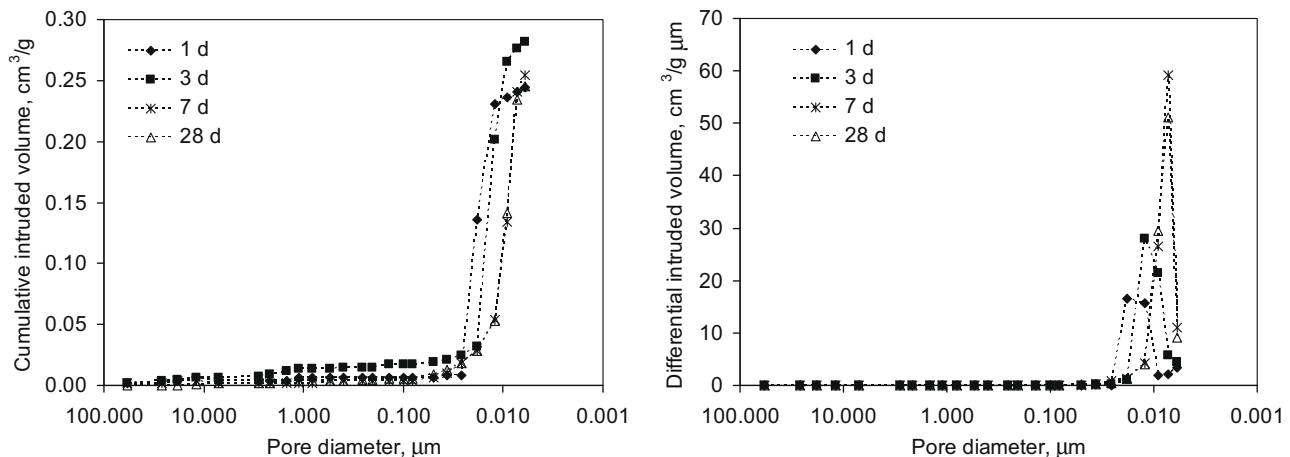


Fig. 4. Time development of pore distribution of geopolymer cured at an ambient temperature. The numbers refer to the age of the sample.

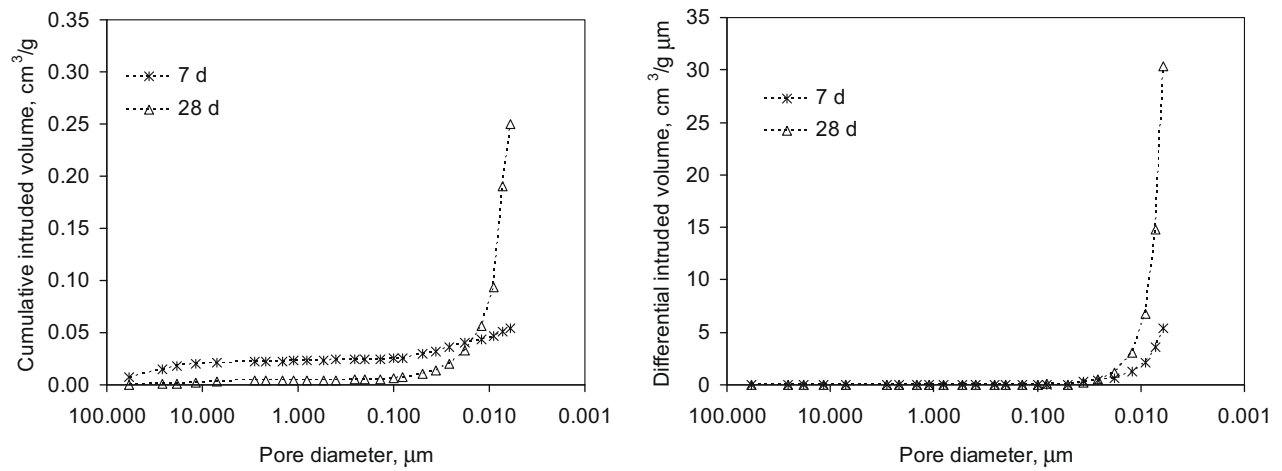


Fig. 5. Time development of pore distribution of geopolymer cured at 10 °C. The numbers refer to the age of the sample.

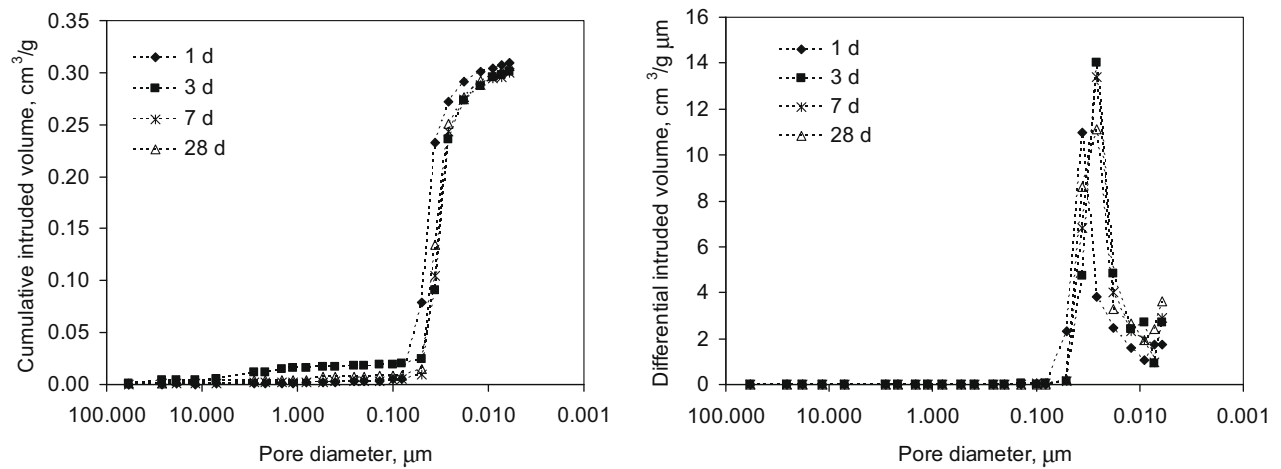


Fig. 6. Time development of pore distribution of geopolymer cured at 80 °C for 4 h. The numbers refer to the age of the sample.

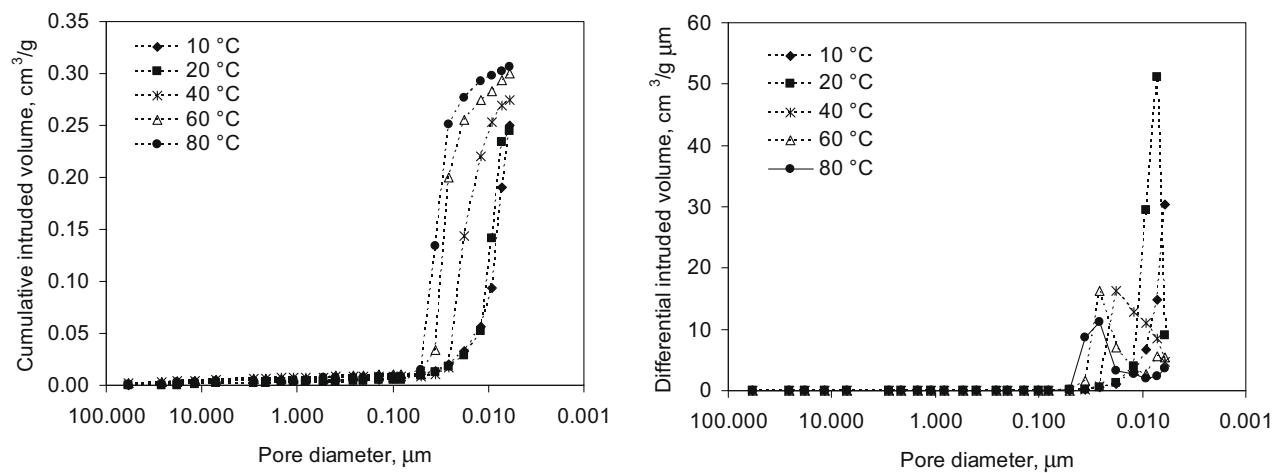


Fig. 7. Comparison of pore distribution of geopolymers cured at 10, 20, 40, 60, and 80 °C at the age of 28 days. Curing at 40, 60 and 80 °C was carried out for initial 4 h.

to EN 196-1 standard [26]. Pore distribution was evaluated by means of mercury intrusion porosimetry analysis that was conducted on paste samples using Micromeritics Poresizer 9300 porosimeter that can generate a maximum pressure of 207 MPa and can evaluate a theoretical pore diameter of 0.006 μm . The MIP test is performed in two steps: the low pressure step first evacuates gases, fills the sample holder with mercury and performs porosimetry from about 7 to 179 kPa; the high pressure step reaches pressures between 414 kPa and 207 MPa. The contact angle and surface tension assumed for all tests were 130° and 485 dyn/cm, respectively. FTIR spectra of the finely ground samples were taken using Nicolet 380 FTIR spectrometer, with a single reflectance silicon ATR attachment. The FTIR spectra were collected in the absorbance mode from 400 to 4000 cm^{-1} at resolution of 1 cm^{-1} and 64 scans per spectrum. All geopolymer samples were analyzed for the position of the main Si–O–T (T=Si or Al) asymmetric stretching band.

3. Results and discussion

3.1. Mechanical properties

Almost all tested specimens set and formed hard structure within 24 h after preparation except for the specimens that were cured under low temperature conditions. Temperature of 10 °C retarded the setting and hardening of geopolymer mixture to 4 days, hence the flexural and compressive strengths could not be collected within this period. Bulk density of hardened samples as a function of curing temperature is presented in Fig. 1. The values slightly decrease with rising temperature which can indicate that higher curing temperature makes the hardened structure less dense, and hence less compact.

Fig. 2 represents the influence of curing temperature on the compressive and flexural strengths of geopolymer mortar at the age of 1, 3, 7 and 28 days after mixing. The reference mortar cured at an ambient temperature reached the compressive strength 62 MPa and flexural strength 11.6 MPa at the age of 28 days. The early-age strengths of geopolymer cured at 10 °C are equal to zero due to retarded setting of geopolymer mixture. As expected, the elevated temperature accelerates the formation of hard structure especially in the early-stage of geopolymerization reaction. Both, compressive and flexural, strengths of geopolymer mortar cured at 60 or 80 °C, respectively, reached their final values just 24 h after mixing and three times exceeded the values observed for samples cured at an ambient temperature. However, the rapid setting prevents the mixture from the formation of more compact and tough structure; hence the compressive strength in 28 days is by 10 MPa lower compared to reference mortar. On the contrary, though the mixture that was cured at decreased temperature exhibits delayed strengths' development, it reached the target value of 62 MPa in 28 days after mixing. The explanation of this behaviour is similar

to the influence of temperature on the strength development of Portland cement [27]. At early ages, the strength increases with the temperature since at higher temperatures the geopolymerization degree is higher, and therefore the amount of reaction products increases. On the other hand, at longer ages, when the geopolymerization degree is approximately the same, the quality of reaction products is the predominant parameter. The geopolymer developed at lower temperature grows slowly and then its quality is better in terms of lower porosity and higher toughness. The flexural strengths of specimens cured at different temperatures show the same trend as for compressive strengths.

The investigation was also focused on the influence of curing time at elevated temperatures. The compressive and flexural strengths of geopolymer as a function of curing time at specified elevated temperature is presented in Fig. 3. Longer curing of geopolymer mixture accelerates the development of strength in the first 24 h of hardening. Meanwhile 1-day compressive strength of geopolymer mortar cured for 1 h at 40 °C was only 13 MPa, the strength increased almost three times to 37 MPa, when such curing was prolonged to 4 h. The final values of strengths were reached approximately in 7 days and they are comparable to those observed for reference sample cured at an ambient temperature (Fig. 2). The influence of curing time on the strength development of geopolymer mortars treated at 60 and 80 °C is very similar. The specimens that were cured for two or more hours at high temperature reached their final strengths in 3 days with the values about 50 MPa. On the contrary, when the mixture was subjected to higher temperature just for 1 h, a trend of the strength development is very similar to that observed with 4-h curing at 40 °C. In this case, the early-age compressive strength is approximately 30 MPa but moreover, the final strength is not decreased.

3.2. Pore structure analysis

The mercury intrusion porosimetry test was carried out to determine the pore distribution in geopolymer matrix. Development of pore structure in terms of cumulative and differential pore volumes for specimens cured at 10 and 20 °C is presented in Figs. 4 and 5. The maximum of pores in geopolymer that was treated at an ambient temperature lies between 7 and 20 nm in diameter. At early ages, the pores are slightly larger, but during aging the maximum volume shifts towards smaller pores and these changes in pore structure are practically finished in 7 days. This shifting of mean pore size can be explained by gradual filling of larger pores with reaction products as the geopolymerization proceeds.

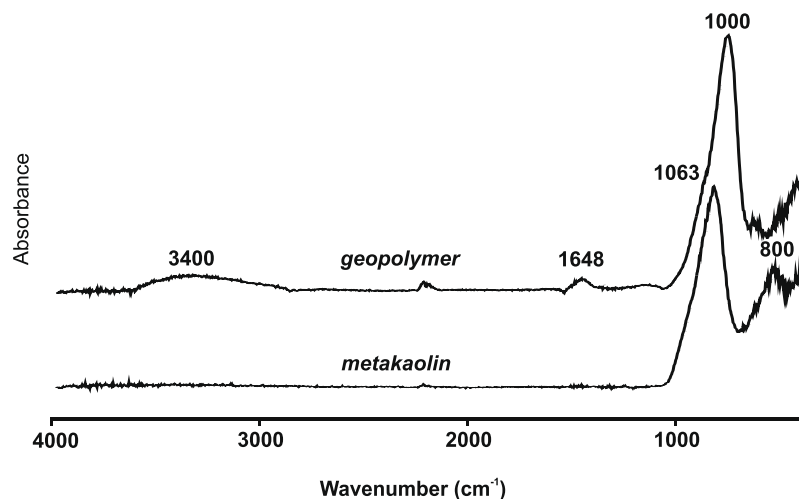


Fig. 8. FTIR spectra for unreacted metakaolin and metakaolin-based geopolymer at the age of 28 days.

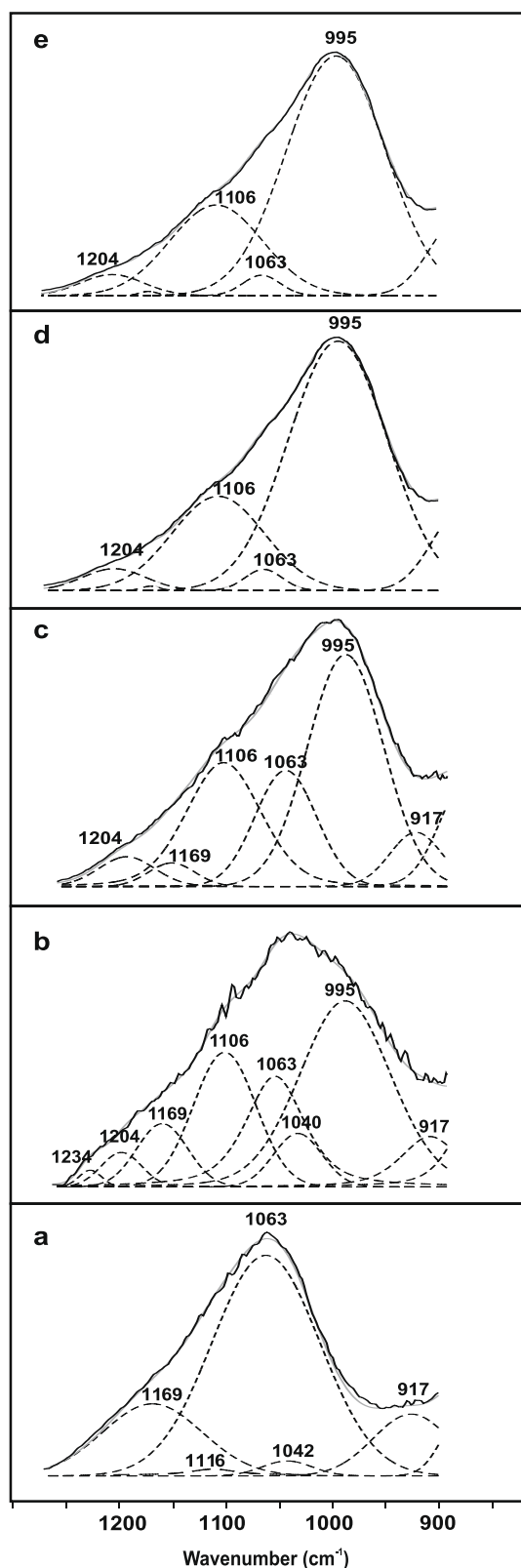


Fig. 9. Deconvolution of the "main band" associated with Si–O–T stretching mode in metakaolin and hardened geopolymer paste cured at 10 °C at different reaction times: (a) unreacted metakaolin; (b) 1 day; (c) 3 days; (d) 7 days; (e) 28 days after mixing.

The early age pore distribution of geopolymer treated at 10 °C could not be measured due to delayed setting, and therefore only pore distribution curves for 7 and 28 days of hardening are pre-

sented. Although the cumulative intruded volume after 7 days seems to be unreasonably small, it probably does not correspond to the total volume of pores presented in the paste. This discrepancy might have been caused by some mixing water that remained in very small pores even after vacuum drying due to strong capillary force. Increased temperature during drying would probably help to remove all absorbed free water, but unfortunately it would also accelerate the geopolymerization reaction, and therefore such possibility has been avoided. The picture shows that some large pores occurred in the freshly formed basic hard structure, nevertheless after 28 days only very few pores, smaller than 40 nm, remained in the structure of geopolymer matrix, and the pore distribution was very similar to that of the reference sample.

Fig. 6 represents development of pores in specimens that were cured at 80 °C for 4 h. At higher temperatures, the final pore structure is almost built within the first 24 h. The maximum volume was observed for pores of the size between 20 and 50 nm and it practically did not change during aging of geopolymer. Larger pores observed in heated geopolymer, compared to the reference matter, have their origin in rapid formation of hard structure. When the process of hardening proceeds too quickly, it results in less ordered structure of poorer quality with relatively larger pores left in the matrix, whereas in the geopolymer cured at lower temperatures the products of geopolymerization can gradually fill the voids in the basic structure and thus make it denser. This suggestion is supported by comparison of pore distribution in geopolymers cured at different temperatures (Fig. 7). The pore volume slightly increases with rising temperature of curing. Therefore curing at higher temperatures results in increased pore volume and pore size, whereas low temperature curing increases the bulk density. Cumulative pore volume of geopolymer is generally twice as high in comparison with materials based on Portland cement or alkali activated slag [28]. This quality is also reflected in higher permeability and absorption capacity of water and aqueous solutions [29]. The results from porosimetry measurements also correspond very well with development of compressive and flexural strengths discussed above.

3.3. Infrared spectrometric analysis

FTIR spectra of metakaolin geopolymer were analyzed generally for the asymmetric stretching vibration of Si–O–T linkages that were estimated to be represented by the point of maximum absorbance in the region 1300–900 cm^{-1} ("main band"). Other major bands observed in the spectra were assigned to stretching (broad band at 3400 cm^{-1}) and bending (1648 cm^{-1}) vibration of absorbed H_2O (Fig. 8). Absorption features in the inspected region of the spectra are broadened and asymmetric and imply the overlap of more bands. Deconvolution of the main band of hardened geopolymer cured at decreased temperature revealed the overlap of several fundamental bands (Fig. 9). The fit was performed with Gaussian functions and the regression coefficient (r^2) varied between 0.9862 and 0.9996. The bands at 1169 and 1063 cm^{-1} are associated with asymmetric stretching mode of original metakaolin and they were found in all spectra. Observing this feature even after 28 days means that a part of unreacted metakaolin still remains in hardened material. Very weak bands at 1116 and 1042 cm^{-1} in metakaolin spectrum give the evidence of little uncalcined kaolin in starting material [30]. The most intensive band at 995 cm^{-1} represents asymmetric vibration of oxygen linkages between tetrahedra in geopolymeric structure, whereas the band at 1106 cm^{-1} is assigned to stretching mode of internal tetrahedra. Weak band at 1204 cm^{-1} is connected with stretching vibrations of terminal Si–OH groups in partially hydrated geopolymer matter [31]. Band at 917 cm^{-1} that appears in spectrum of metakaolin and diminishes during geopolymerization is associated

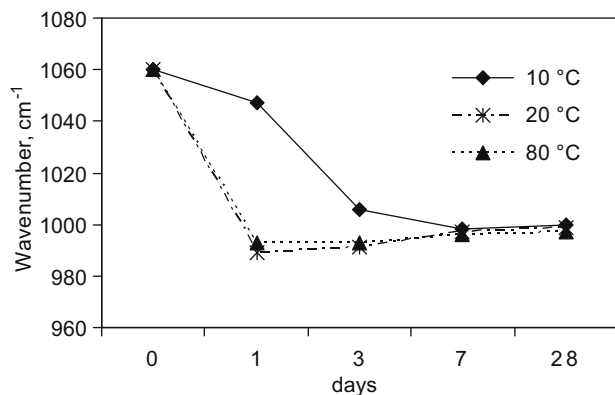


Fig. 10. Shift over time in the position of Si–O–T asymmetric stretching band for geopolymers cured at different temperatures.

with symmetric Al–O–Si stretching vibrations of metakaolin. The spectrum of geopolymer after 1 day (9b) contains also two bands with wavenumbers 1234 and 1042 cm^{-1} . These bands probably represent asymmetric TO_4 stretching vibrations of water glass activator and some intermediate species because they completely disappear during setting process. Broad band with low intensity at 800 cm^{-1} that can be clearly observed in the spectrum of pure metakaolin (Fig. 8) belongs to O–Al–O bending vibrations of AlO_4 tetrahedra [15]. The bands that correspond with other bending modes of TO_4 units could not be resolved due to considerable signal noise below 800 cm^{-1} associated with ATR technique.

The reaction course and formation of hard structure can be described by analyzing a shift of the main band that is clearly visible from the comparison of IR spectra recorded at different reaction times for the mixture that was cured at 10 °C. As the geopolymerization reaction proceeds the intensity of band at 1063 cm^{-1} drops and band at 995 cm^{-1} associated with new phase rises which is connected with a shift of the overlapping peak towards lower frequencies. Fig. 10 shows the shift of the main band with time for three different curing temperatures (10, 20 and 80 °C). The mixtures that were treated at an ambient or higher temperature exhibit full shift within the first 24 h of hardening process, whereas in case of specimens cured at low temperature the final position of the main band was observed in 7 days. This observation is consistent with setting time and formation of basic hard structure. The setting time for cooled specimens was estimated to be 4 days whereas thermally treated mixtures set within 2–4 h.

Although MAS NMR techniques provide more detailed view of geopolymerization process [20], FTIR spectroscopy is, however, a possible and simple method for the monitoring of geopolymerization reaction and formation of the hard structure. Analysis of FTIR spectra in the region of Si–O–T stretching vibrations revealed residues of unreacted metakaolin in hardened geopolymer matrix regardless of curing regime. It supports a hypothesis that metakaolin grains are not fully dissolved during geopolymerization process but the reaction takes place in the surface layer of the solid particles [20]. The reaction rate is then controlled mainly by diffusion of hydroxide and silicate ions through primary geopolymer gel which is affected mainly by curing temperature at the early-stage of the reaction.

4. Conclusions

In this study, the effect of curing temperature and time of curing at elevated temperatures on the mechanical properties of metakaolin-based geopolymer was investigated by measuring compressive and flexural strengths' development over time. The obtained

results were explained on the basis of pore distribution analysis carried out using mercury intrusion porosimetry equipment and the microstructure changes on molecular level were examined by means of FTIR spectroscopy. The experimental and analytical studies led to the following conclusions.

1. Curing temperature has an essential effect on setting and hardening of metakaolin-based geopolymer. At an ambient and elevated temperature the matter set practically within 4 h at the latest. On the contrary, the setting was delayed to about 4 days when the mixture was treated at temperature as low as 10 °C, but it had no negative effect on the quality and properties of hardened product at the age of 28 days. Evaluated mechanical tests showed that both early-age and final mechanical properties of geopolymer material are greatly dependent on curing temperature. Higher curing temperatures increase the early-age compressive and flexural strengths, which can even reach their target values within 1 day. However, the 28 days strengths were markedly lower in comparison with those observed for specimens that were treated at an ambient or decreased temperature because the quick formation of the hard structure probably does not result in such a good quality product. Moreover, the effect of temperature depends on curing time. Curing for just 1 h at an elevated temperature did not cause remarkable changes in strengths development but longer treatment was responsible for a considerable acceleration of the reaction rate and increase in early-age strengths.
2. Porosimetry measurements showed that the mean pore size slightly decreased with aging, and this was true for all specimens treated in different curing conditions. However, main differences in pore distribution appeared between samples cured at different temperatures. Elevated temperature during early-stage of hardening process led to the formation of larger pores and increase in cumulative pore volume, which has a negative effect on the final mechanical properties of geopolymer material.

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References

- [1] Davidovits J. Mineral polymers and methods of making them. US Patent 4349386; 1984.
- [2] Davidovits J. 30 years of successes and failures in geopolymer applications. Market trends and potential breakthroughs. In: Proceedings of international conference "Geopolymer 2002", Melbourne, 2002. p. 1–16.
- [3] Duxson P, Fernández-Jiménez A, Provis JL, Luckey GC, Palomo A, van Deventer JSJ. Geopolymer technology: the current state of the art. *J Mater Sci* 2007;42(9):2917–33.
- [4] Davidovits J. Geopolymers: inorganic polymeric new material. *J Therm Anal* 1991;37(8):1633–56.
- [5] Lyon E, Balaguru PN, Foden A, Sorathia U, Davidovits J, Davidovits M. Fire resistant aluminosilicate composites. *Fire Mater* 1997;21(2):67–73.
- [6] Xu H, van Deventer JSJ. Geopolymerisation of multiple minerals. *Miner Eng* 2002;15(12):1131–9.
- [7] Gordon M, Bell JL, Kriven WM. Comparison of naturally and synthetically-derived, potassium-based geopolymers. *Ceram Trans* 2005;165:95–106.
- [8] Xu H, van Deventer JSJ. The geopolymerisation of aluminosilicate minerals. *Int J Miner Proc* 2000;59(3):247–66.
- [9] Davidovits J. Properties of geopolymer cements. In: First international conference of alkaline cements and concretes, Kiev, 1994. p. 131–49.
- [10] Weng L, Sagoe-Crentsil K. Dissolution processes, hydrolysis and condensation reactions during geopolymer synthesis: part I—low Si/Al ratio systems. *J Mater Sci* 2007;42(9):2997–3006.

- [11] Sagoe-Crentsil K, Weng L. Dissolution processes, hydrolysis and condensation reactions during geopolymer synthesis: part II—low Si /Al ratio systems. *J Mater Sci* 2007;42(9):3007–14.
- [12] Walton RI, Millange F, O'Hare D, Davies AT, Sankar G, Catlow CRA. An in situ energy-dispersive X-ray diffraction study of the hydrothermal crystallization of zeolite – a. 1: influence of reaction conditions and transformation into sodalite. *J Phys Chem B* 2001;105(1):83–90.
- [13] Criado M, Fernández-Jiménez A, de la Torre AG, Aranda MAG, Palomo A. An XRD study of the effect of the $\text{SiO}_2/\text{Na}_2\text{O}$ ratio on the alkali activation of fly ash. *Cem Concr Res* 2007;37(5):671–9.
- [14] Alonso S, Palomo A. Calorimetric study of alkaline activation of calcium hydroxide–metakaolin solid mixtures. *Cem Concr Res* 2001;31(1):25–30.
- [15] Granizo ML, Blanco-Varela MT, Palomo A. Influence of the starting kaolin on alkali-activated materials based on metakaolin. Study of the reaction parameters by isothermal conduction calorimetry. *J Mater Sci* 2000;35(24):6309–15.
- [16] Rahier H, van Mele B, Biesemans M, Wastiels J, Wu X. Low-temperature synthesized aluminosilicate glasses. *J Mater Sci* 1996;31(1):71–9.
- [17] Rahier H, Simons W, van Mele B, Wastiels J. Low-temperature synthesized aluminosilicate glasses: part III Influence of the composition of the silicate solution on production, structure and properties. *J Mater Sci* 1997;32(9):2237–47.
- [18] Rahier H, Wastiels J, Biesemans M, Willem R, van Assche G, van Mele B. Reaction mechanism, kinetics and high temperature transformations of geopolymers. *J Mater Sci* 2007;42(9):2982–96.
- [19] Palomo A, Blanco-Varela M, Granizo M, Puertas F, Vazquez T, Grutzeck M. Chemical stability of cementitious materials based on metakaolin. *Cem Concr Res* 1999;29(7):997–1004.
- [20] Barbosa VFF, Mackenzie KJD, Thaumaturgo C. Synthesis and characterisation of materials based on inorganic polymers of alumina and silica: sodium polysialate polymers. *Int J Inorg Mater* 2000;2(4):309–17.
- [21] Barbosa VFF, Mackenzie KJD. Synthesis and thermal behaviour of potassium sialate geopolymers. *Mater Lett* 2003;57(9):1477–82.
- [22] Yunsheng Z, Wei S, Zongjin L. Preparation and microstructure of K-PSDS geopolymeric binder. *Colloids Surfaces A: Physicochem Eng Aspects* 2007;302:473–82.
- [23] Granizo ML, Blanco-Varela MT, Martínez-Ramírez S. Alkali activation of metakaolins: parameters affecting mechanical, structural and microstructural properties. *J Mater Sci* 2007;42(9):2934–43.
- [24] De Silva P, Sagoe-Crensil K, Sirivivatnanon V. Kinetics of geopolymerization: role of Al_2O_3 and SiO_2 . *Cem Concr Res* 2007;37(4):512–8.
- [25] Brunauer S, Emmett PH, Teller E. Adsorption of gases in multimolecular layers. *J Am Chem Soc* 1938;60(2):309–19.
- [26] EN 196-1:2005. Methods of testing cement – part 1: determination of strength.
- [27] Collepardi M. The new concrete. Italy: Grafiche Tintoretto; 2006. p. 221–4.
- [28] Rovnaník P, Bayer P, Rovnaníková P. Utilisation of alkali activated aluminosilicates as fire protecting materials. In: Concrete for fire engineering, proceedings of 7th international congress – concrete: construction's sustainable option, Dundee, UK; 2008. p. 273–82.
- [29] Okada K, Ooyama A, Isobe T, Kameshima Y, Nakajima A, MacKenzie KJD. Water retention properties of porous geopolymers for use in cooling applications. *J Eur Ceram Soc* 2009;29(10):1917–23.
- [30] Akolekar D, Chaffee A, Russell FH. The transformation of kaolin to low-silica X zeolite. *Zeolites* 1997;19(5):359–65.
- [31] Benesi HA, Jones AC. An infrared study of the water–silica gel system. *J Phys Chem* 1959;63(2):179–82.